

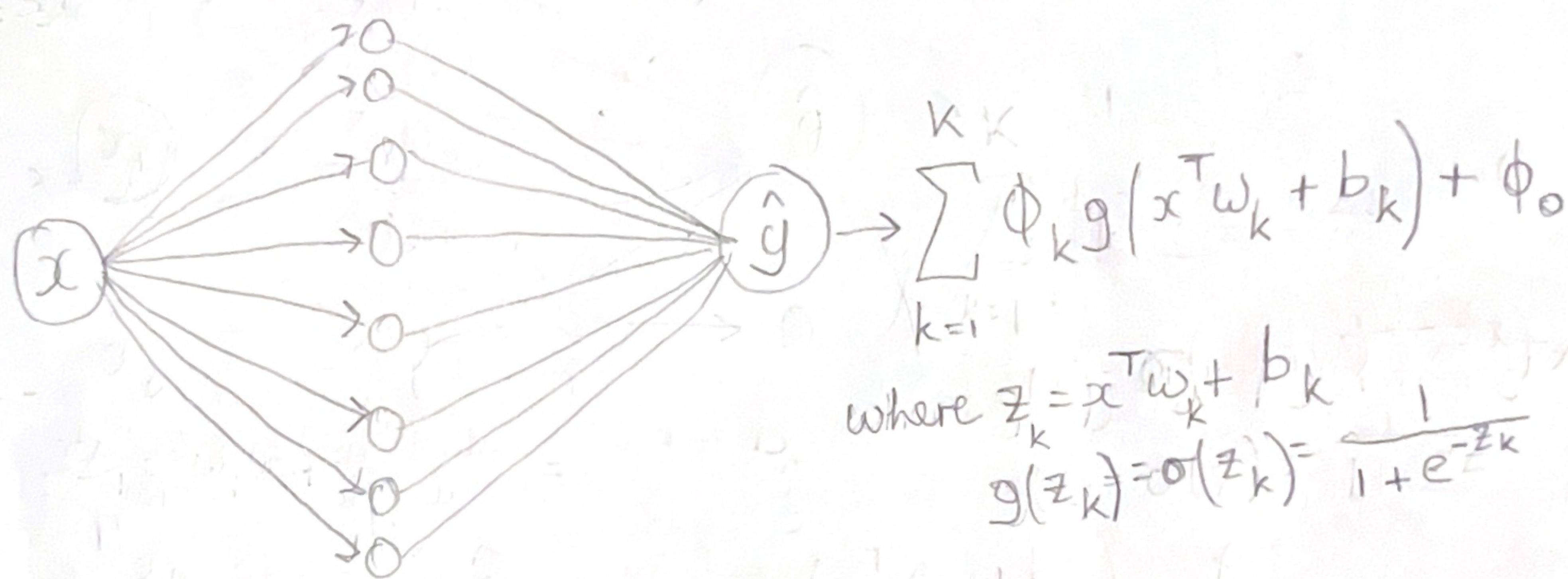
Problem 5 - Approximating a Function with Pulses

a) Pulse 1: $(s_1, s_2, h) = (0, 1, 3)$

Pulse 2: $(s_1, s_2, h) = (1, 2, 4)$

Pulse 3: $(s_1, s_2, h) = (2, 3, 4)$

Pulse 4: $(s_1, s_2, h) = (3, 4, 3)$



- On interval $[0, 1]$; pulse 1:
 x near 0 $\rightarrow \omega_1 = 0, b_1 = 1 \times 10^9, g(z_1) \approx 1, \phi_1 = 3$
 x near 1 from left $\rightarrow \omega_2 = -1 \times 10^9, b_2 = 0, g(z_2) \approx 0, \phi_2 = 0$
- On interval $[1, 2]$; pulse 2:
 x near 1 from right $\rightarrow \omega_3 = 1 \times 10^9, b_3 = 0, g(z_3) \approx 1, \phi_3 = 4$
 x near 2 from left $\rightarrow \omega_4 = -1 \times 10^9, b_4 = 0, g(z_4) \approx 0, \phi_4 = 0$
- On interval $[2, 3]$; pulse 3:
 x near 2 from right $\rightarrow \omega_5 = 1 \times 10^9, b_5 = 0, g(z_5) \approx 1, \phi_5 = 4$
 x near 3 from left $\rightarrow \omega_6 = -1 \times 10^9, b_6 = 0, g(z_6) \approx 0, \phi_6 = 0$
- On interval $[3, 4]$; pulse 4:
 x near 3 from left $\rightarrow \omega_7 = 1 \times 10^9, b_7 = 0, g(z_7) \approx 1, \phi_7 = 3$
 x near 4 from right $\rightarrow \omega_8 = -1 \times 10^9, b_8 = 0, g(z_8) \approx 0, \phi_8 = 0$
- $\phi_0 = 0$